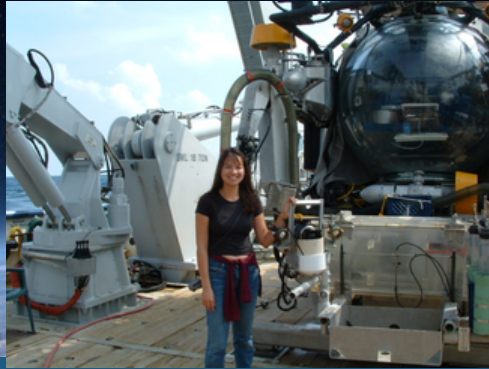


What does methane tell us about life on Mars Earth?

Seeps, Salt marshes & Serpentinities



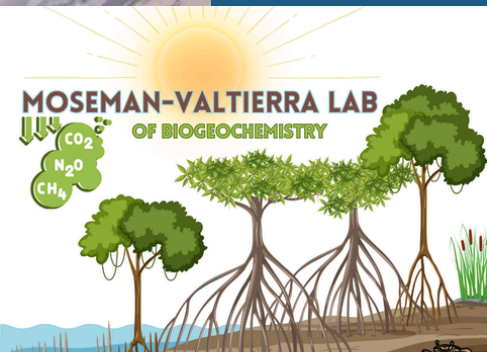
Serena Moseman-Valtierra

Department of Biological Sciences, URI

Oct 23, 2026

3 p.m.

CBLS 010
CELS Colloquium



Methane is a small but potent greenhouse gas that is skyrocketing in Earth's atmosphere. While detection of methane on Mars is sparking excitement about potential extraterrestrial life, I am curious about methane here at home. In Earth's largest natural methane source- wetlands- gas emissions reflect dynamic microbial responses to changing environments. Methane emissions from tidal wetlands, terrestrial and marine seeps, need to be better constrained. In this talk I will present findings from first synthesis of methane data from North American tidal marshes. Also, I will share preliminary measurements of abiotic methane emissions from subterranean ophiolites of Northern California. There, at the Coastal Range Ophiolite Microbial Observatory, our recent analyses suggest that microbes oxidize methane in alkaline groundwaters. Overall, our findings assert the need for integrated measurement approaches of methane in dynamic ecosystems. Further, predictions of future methane levels on Earth must consider potential shifts in both sinks and sources as microbial communities evolve.